Experiment 4: Thermochemistry in Action

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CHEM 110: Integrated Science – Chemistry

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1. Introduction

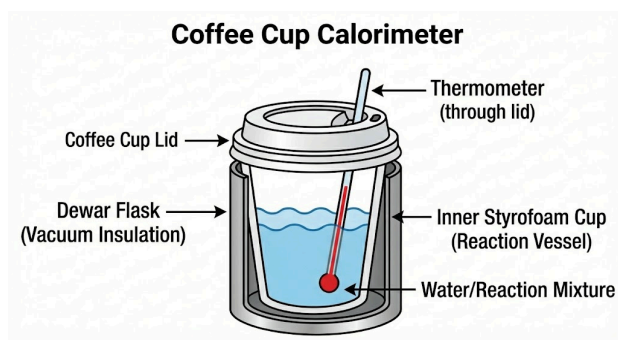
When salts dissolve in water, their ionic bonds are broken and replaced by ion–dipole attractions between the ions and surrounding water molecules. This process can either absorb or release energy, depending on the balance between bond breaking and bond formation. By utilizing this energy change, hot and cold packs can be made. However, since different salts produce different magnitudes and directions of heat transfer, an optimal salt needs to be selected for maximizing thermal efficiency. The heat exchanged during dissolution can be quantified using the equation $q = ms\Delta T$, where m is the mass of the solution, s is the specific heat capacity of water, and ΔT is the observed temperature change. According to the First Law of Thermodynamics, energy cannot be created or destroyed; therefore, any heat absorbed or released during salt dissolution must be exchanged with the surroundings. Calorimetry allows this heat transfer to be measured experimentally by relating the heat of the system to that of the surroundings through the relationship $q_{system} = -q_{surroundings}$. By applying this principle, the thermal behavior of different salts can be directly compared and evaluated, and thus an optimal hot/cold pack can be created.

2. Procedure

2.1. Calorimetry

A simple calorimeter was built by inserting a coffee cup into a Dewar flask. The coffee cup was filled with deionized water, and the temperature was measured using a thermometer shown in Figure 1. Calorimetry was performed by recording the initial temperature, adding salt to the deionized water, and recording the resulting final temperature. This process was repeated to verify that the temperature change fell within a targeted 3-8 °C with a variation of less than 0.5°C.

Figure 1: Diagram of Coffee Cup Calorimeter



2.2. Cold Pack Design

After the optimization of the salt mass and type, a prototype pack was created. This pack has three main goals: be water-tight, have a temperature change of between 3-8°C, and have a way to introduce the salt when desired. After the prototype was made and tested to verify that it

met these conditions, two consumer packs were created and exchanged with another group for consumer testing. Finally, any critical suggestions were implemented for the final product.

3. Results and Discussion

3.1. Initial Optimization and Testing

The enthalpy changes of solution for various potential compounds were calculated to determine the most efficient endothermic agents listed in Table 1. Based on molar enthalpy, the three most promising candidates were Lithium Chloride (LiCl), Calcium Chloride (CaCl2), and Potassium Chloride (KCl).

Table 1: Enthalpy change of solution per yuan of selected compounds

Compound name	Chemical formula	Molar $\Delta H_{solution}$ (kJ/mol)	Molecular weight (g/mol)	$\Delta H_{solution}$ (kJ/g)	Price (yuan/g)	ΔH solution (kJ/yuan)	Mass needed (g) for 100 g of water	Price (yuan) for the mass needed
Magnesium sulfate	$MgSO_4$	-91.2	120.36	-0.7577	4	-0.1894	3.31	13.2
Calcium chloride	$CaCl_2$	-82.9	110.98	-0.747	4	-0.1868	3.36	13.4
Lithium chloride	LiCl	-37	42.39	-0.8728	6	-0.1455	2.88	17.3
Sodium acetate	$C_2H_3NaO_2$	-19.7	82.03	-0.2402	4	-0.0601	10.4	41.6
Potassium acetate	$KC_2H_3O_2$	-15.3	98.14	-0.1559	4	-0.039	16.1	64.4
Sodium sulfate	Na_2SO_4	-2.3	142.04	-0.0162	4	-0.0041	155	620
Glucose	$C_6H_{12}O_6$	10.5	180.16	0.0583	3	0.0194	43.1	129

Potassium iodide	KI	20.3	166	0.1223	4	0.0306	20.5	82
Sodium acetate trihydrate	$NaC_2H_3O_2$ *$3H_2O$	19.7	136.08	0.1448	4	0.0362	17.3	69.2
Urea	$CO(NH_2)_2$	14	60.06	0.2331	4	0.0583	10.8	43.2
Ammonium chloride	NH_4Cl	14.8	53.49	0.2767	4	0.0692	9.07	36.3
Potassium chloride	KCl	17.2	74.55	0.2307	3	0.0769	10.9	32.7

The objective was to design the most efficient thermal pack, with the flexibility to create either a hot or cold pack based on performance. Calcium Chloride (CaCl2), and Lithium Chloride (LiCl) were identified as candidates for heating, and Potassium Chloride (KCl) for cooling.

- Calcium Chloride: Theoretically, this was the best heating agent. However, during preliminary testing, the pellets exhibited poor solubility, dissolving too slowly to generate an immediate or sustained temperature spike.
- Lithium Chloride (LiCl): This salt is also highly exothermic and appears to be a viable option. However, due to its high cost and toxicity, it was excluded from selection.
- Potassium Chloride (KCl): A cold pack design was subsequently adopted. Unlike the calcium pellets, the powder dissolved rapidly and completely, creating an immediate cooling effect. Therefore, it was selected for use in the final product.

Table 2 shows the testing data using calorimetry. Using KCl, an average temperature decrease of 4.1°C was achieved, which fell within the target range of 3–8°C with a <5°C variance. Because of this favorable data, it was decided to proceed with the use of potassium chloride for the cold pack prototype.

Table 2: Calorimetry Optimization Data Using Potassium Chloride

Trial	Mass (g)	Water Vol. (mL)	$T_{initial}$ (°C)	T_{final} (°C)	ΔT (°C)
1	5.00	50	23.0	19.1	3.9
2	5.00	50	23.2	18.9	4.3

3.2. Cold Pack Prototype Testing

For the prototype, three requirements were established. First, a temperature range of 3–8°C with a variance of <5°C was required; second, it was necessary to ensure the bag did not leak; third, a method to separate the salt and water was needed so that the salt could be introduced to the water when desired. In the initial prototype, a baggie clip was placed at the top of a zip bag, directly under the zipper seal. The salt sat freely between the baggie clip and the zipper seal. While free-floating, issues with salt leakage were observed, and since the water volume shifted, difficulties in measuring the temperature were encountered. To solve this, the salt was placed into a centrifuge tube, and the baggie clip was moved down; the final prototype is shown in Figure 2. After the design was finalized, testing was initiated to ensure requirements were met. Upon verification of the prototype, the data for which is shown in Table 3, two additional packs were constructed for consumer testing.

Figure 2: Cold Pack Prototype Design

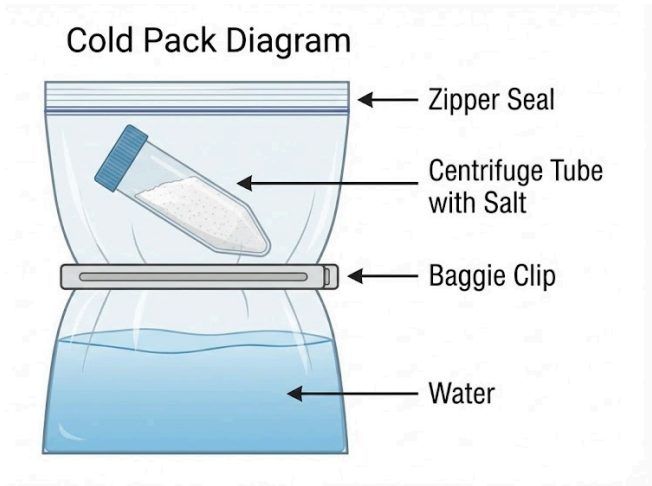


Table 3: Prototype Cold Pack Performance Data

Trial	Mass (g)	Water Vol. (mL)	$T_{initial}$ (°C)	T_{final} (°C)	ΔT (°C)
1	5.00	50	25.7	21.3	3.4
2	5.00	50	25.0	20.9	4.0

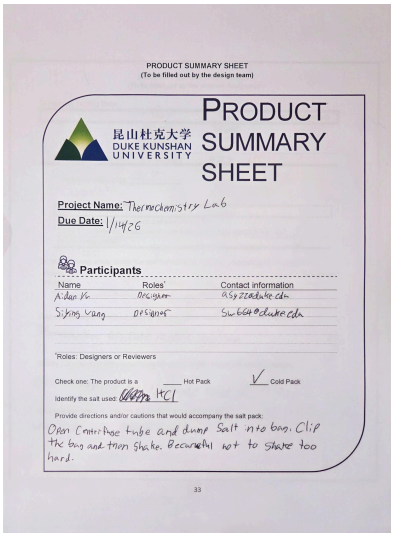
3.3. Product Consumer Testing

Two sample products were made based on our successful prototype and were prepared for consumer testing. Figure 5 shows the testing data of our product evaluation sheet that we filled out, which was given to the testing group for reference. Figure 4 shows the tester's evaluation of our product. Our feedback showed that our product was good and that no iterations were needed. Figure 3 is a summary of our product, which can be given to the consumer for reference.

3.4. Possible Improved Design of Hot/Cold Pack

Based on designer feedback shown in Figure 5, it was noted that the "temp change could be better." A potential modification to obtain a stronger cooling effect would be to increase the mass of Potassium Chloride from 5.0 to 7.5 grams while maintaining the same water volume. This was never implemented because it was not deemed necessary per the instructions of the lab.

Figure 3: Product Summary Sheet



PRODUCT SUMMARY SHEET
(To be filled out by the design team)

昆山杜克大学 DUKE KUNSHAN UNIVERSITY

PRODUCT SUMMARY SHEET

Project Name: Thermochemistry Lab
Due Date: 1/14/26

Participants

Name	Roles*	Contact information
Aidan Xi	Designer	ax224@duke.edu
Siking Wang	Designer	sw664@duke.edu

*Roles: Designers or Reviewers

Check one: The product is a ___ Hot Pack ✓ Cold Pack

Identify the salt used: ~~NaCl~~ KCl

Provide directions and/or cautions that would accompany the salt pack:
Open Centrifuge tube and dump Salt into bag. Clip the bag and then shake. Becareful not to shake too hard.

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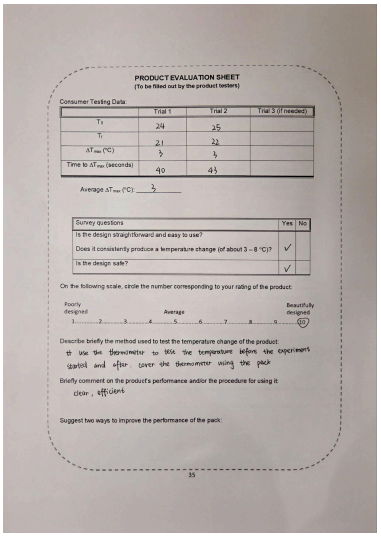
Figure 4: Product Tester Evaluation Sheet



PRODUCT EVALUATION SHEET
(To be filled out by the product testers)

Consumer Testing Data:

	Trial 1	Trial 2	Trial 3 (if needed)
T_0	24	25	
T_f	21	22	
ΔT_{max} (°C)	3	3	
Time to ΔT_{max} (seconds)	40	43	

Average ΔT_{max} (°C): 3

Survey questions	Yes	No
Is the design straightforward and easy to use?		
Does it consistently produce a temperature change (of about 3 – 8 °C)?	✓	
Is the design safe?	✓	

On the following scale, circle the number corresponding to your rating of the product:

Poorly designed 1 2 3 4 5 6 7 8 9 (10) Beautifully designed

Describe briefly the method used to test the temperature change of the product:
It use the thermometer to test the temperature before the experiment started and after. cover the thermometer using the pack

Briefly comment on the product's performance and/or the procedure for using it
clear, efficient

Suggest two ways to improve the performance of the pack:

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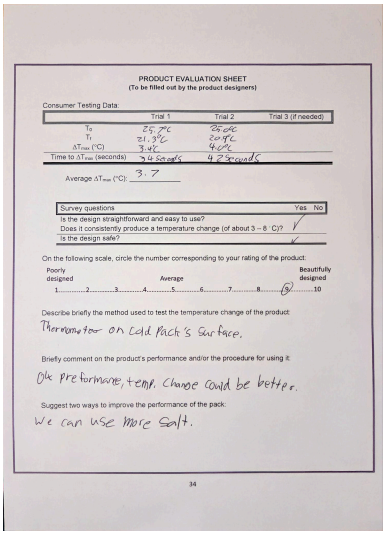
Figure 5: Product Designer Evaluation Sheet



PRODUCT EVALUATION SHEET
(To be filled out by the product designers)

Consumer Testing Data:

	Trial 1	Trial 2	Trial 3 (if needed)
T_0	25.7°C	20.0°C	
T_f	21.3°C	20.9°C	
ΔT_{max} (°C)	3.4°C	4.0°C	
Time to ΔT_{max} (seconds)	34 seconds	42 seconds	

Average ΔT_{max} (°C): 3.7

Survey questions	Yes	No
Is the design straightforward and easy to use?		
Does it consistently produce a temperature change (of about 3 – 8 °C)?	✓	
Is the design safe?	✓	

On the following scale, circle the number corresponding to your rating of the product:

Poorly designed 1 2 3 4 5 6 7 8 (9) 10 Beautifully designed

Describe briefly the method used to test the temperature change of the product:
Thermometer on Cold Pack's Surface.

Briefly comment on the product's performance and/or the procedure for using it:
Ok preformance, temp. change could be better.

Suggest two ways to improve the performance of the pack:
We can use more Salt.

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3.5. Cost Breakdown

The commercial feasibility of the design was also evaluated based on the cost per unit. The cost was determined to be 28.60 RMB per pack as shown in Table 4.

Table 4: Total Cost of Designed Cold Pack

Material	Quantity	Unit Cost	Total (RMB)
Salt(KCl)	5.00 g	3.00 RMB/g	15.00
DI Water	50 mL	25 Cents/mL	12.5
Snack Size Baggie	1	0.60 RMB	0.60
Centrifuge Tube	1	0.50 RMB	0.50
Total			**28.60**

4. Summary and Reflection

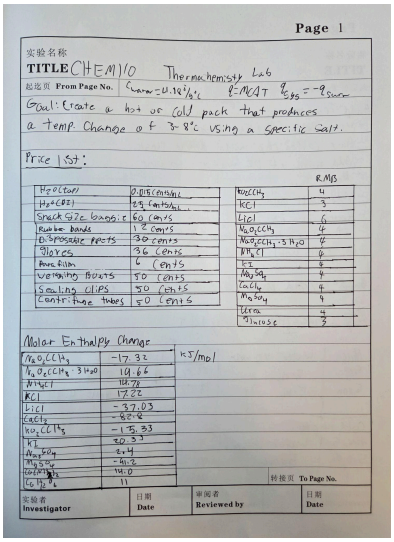
This experiment successfully demonstrated the use of Potassium Chloride within a cold pack design. While $CaCl_2$ offered a higher theoretical heat exchange, the practical constraints of solubility led to the selection of KCl, which provided a reliable temperature drop of ≈ 4.0°C. The importance of balancing theoretical thermodynamic values with physical properties, such as solubility kinetics, when engineering consumer hot/cold packs was highlighted by this iterative process.

5. Appendix

Figure 6: Raw Laboratory Notebook Data (Page 1)

Figure 7: Raw Laboratory Notebook Data (Page 2)

Figure 8: Raw Laboratory Notebook Data (Page 3)



Page 1

実験名称
TITLE CHEM110 Thermochemistry Lab

起迄頁 From Page No. $C_{barrow} = 0.182\ J/g°C$ $q = mC\Delta T$ $q_{gas} = -q_{solution}$

Goal: Create a hot or cold pack that produces a temp change of 3-8°C using a specific salt.

Price list:

H_2O (tap)	0.015 cents/mL
H_2O (DI)	25 cents/mL
Snack Size bags	60 cents
Rubber Bands	18 cents
disposable pipets	30 cents
Gloves	36 cents
Parafilm	6 cents
Weighing Boats	50 cents
Sealing Clips	50 cents
Centrifuge tubes	50 cents

R.MB

KO_2CCH_3	4
KCl	3
LiCl	6
$NH_4O_2CCH_3$	4
$NaO_2CCH_3 \cdot 3H_2O$	6
NH_4Cl	8
KI	8
Na_2SO_4	4
$CaCl_2$	4
Na_2SO_4	4
Urea	4
Glucose	3

Molar Enthalpy Change

kJ/mol

$Na_2O_2CCH_3$	-17.32
$Na_2O_2CCH_3 \cdot 3H_2O$	19.66
NH_4Cl	15.28
KCl	17.22
LiCl	-37.03
$CaCl_2$	-82.8
KO_2CCH_3	-15.33
KI	20.33
Na_2SO_4	2.4
$MgSO_4$	-91.3
Urea	14.0
$C_6H_{12}O_6$	11

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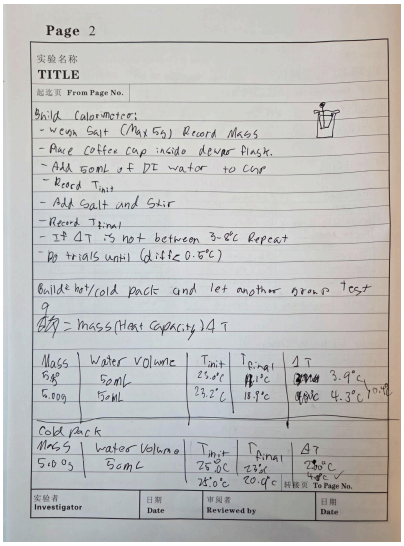

Page 2

実験名称
TITLE

起迄頁 From Page No.

Build Calorimeter:
- Weigh Salt (Max 5g) Record Mass
- Place coffee cup inside dewar flask.
- Add 50mL of DI water to cup
- Record T_{init}
- Add Salt and Stir
- Record T_{final}
- If ΔT is not between 3-8°C Repeat
- Do trials until (diff < 0.5°C)

Build hot/cold pack and let another group test

$q = mass(Heat\ Capacity)\Delta T$

Mass	Water Volume	T_{init}	T_{final}	ΔT
8g	50mL	25.0°C	28.9°C	3.9°C
5.00g	50mL	23.2°C	18.9°C	4.3°C

Cold pack

Mass	Water Volume	T_{init}	T_{final}	ΔT
5.00g	50mL	25.0°C	20.0°C	5.0°C

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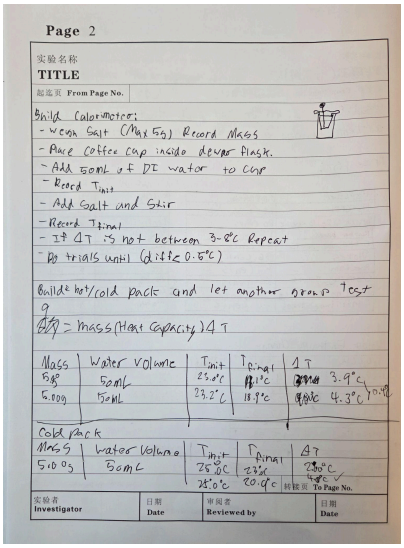

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Cold pack

Mass	Water Volume	T_{init}	T_{final}	ΔT
5.00g	50mL	25.0°C	20.0°C	5.0°C

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